NETFLIX | Global Streaming Intelligence Suite – Watch Behaviour Analysis

Report prepared and analytics conducted by Soham S. Amburle | 16 March 2026

The Watch Behaviour Analysis Dashboard provides a detailed and interactive view of how users engage with content on a simulated global streaming platform inspired by Netflix. Built in Power BI using a synthetic dataset of approximately 2,394 records, the dashboard focuses on analyzing viewing patterns, session behaviour, content consumption, and device usage trends.

This dashboard answers key behavioural questions such as:

- How much time are users spending watching content?
- What is the average watch duration per session?
- What percentage of content is fully completed?
- Which devices are most commonly used for streaming?

How does user engagement vary across genres and content types?

Through KPI cards, charts, maps, and slicers, the dashboard enables users to explore streaming behaviour efficiently and gain insights into platform engagement.

Fig 7.1 Watch Behaviour Analysis

Dashboard Storyline

The dashboard follows a structured analytical flow that helps users understand viewing behavior step by step.

First, the dashboard presents a high-level overview of user engagement through KPI indicators. These metrics provide immediate insights into total watch time, number of sessions, average viewing duration, and completion rates.

Data Disclaimer: This dataset is synthetic and created solely for learning, analysis, and portfolio demonstration. It does not represent real user data or actual activity from Netflix or any streaming platform.

Next, the analysis shifts toward content consumption patterns. This section highlights how users interact with different content types and genres, helping stakeholders understand what drives engagement.

Finally, the dashboard explores device usage and geographic distribution. By analyzing where and how users are watching content, stakeholders can identify key markets and preferred streaming devices.

This layered approach ensures that users can quickly grasp the overall engagement picture while also exploring deeper insights through interactive filters.

Dashboard Elements

The dashboard is composed of three primary analytical components:

- KPI Cards summarizing key engagement metrics
- Interactive Slicers for dynamic filtering
- Visual Charts displaying watch behaviour patterns

Together, these elements create an intuitive and interactive environment for analyzing user engagement.

KPI Cards

Six KPI cards provide a summary of core watch behaviour metrics:

1. **Total Watch Hours**: Total time spent by users watching content on the platform.
2. **Total Watch Sessions**: Total number of viewing sessions recorded.
3. **Average Watch Time**: Average minutes watched per session.
4. **Completion Rate (%)**: Percentage of sessions where content was fully completed.
5. **Average Sessions per User**: Average number of sessions per user.
6. **Average Watch Hours per User**: Average total watch time per user.

These KPIs help measure user engagement, platform usage intensity, and content consumption efficiency.

Slicers

Five slicers allow dynamic filtering of the dashboard:

1. Continent
2. Country
3. Content Type
4. Genre
5. Device Type

These filters enable users to explore viewing patterns across different segments, regions, and content categories.

Main Visualizations

The dashboard includes five primary visuals:

1. **Watch Time by Content Type (Column Chart)**: Shows total watch minutes across movies and series, helping identify which format drives more engagement.
2. **Watch Time by Device Type (Donut Chart)**: Displays the distribution of watch time across devices such as mobile, smart TV, and desktop.

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3. **Average Watch Time by Genre (Bar Chart)**: Highlights which genres have higher engagement based on average watch duration.
4. **Watch Sessions by User (Column Chart)**: Shows distribution of sessions across users, helping identify engagement intensity and user activity patterns.
5. **Watch Hours by Country (Map)**: Visualizes geographic distribution of watch time, identifying regions with higher platform engagement.

These visuals collectively provide a comprehensive understanding of how users interact with the platform across multiple dimensions.

Business Value

This dashboard provides several analytical benefits:

- Helps understand user engagement and viewing intensity
- Identifies content types and genres driving higher watch time
- Analyses device usage patterns for platform optimization
- Highlights geographic trends in content consumption
- Supports data-driven decisions for content strategy and user experience improvements

Overall, the dashboard serves as a behavioural analytics layer, offering actionable insights into how users interact with a streaming platform.

Tools and Data

- Visualization Tool: Power BI
- Dataset: Synthetic global streaming platform dataset
- Total Records: Approximately 2,394 rows across all tables
- Data Model: Snowflake schema
- Tables Used in the Dashboard:
  - Fact Tables
    - Fact_WatchHistory
    - Fact_Subscriptions
    - Fact_Ratings
  - Dimension Tables
    - Dim_Users
    - Dim_Content
    - Dim_Time
    - Dim_Geography
    - Dim_Genre
    - Dim_Language
    - Dim_Device
    - Dim_SubscriptionPlan

The dataset was generated using Mockaroo to simulate typical streaming analytics scenarios such as subscriber distribution, subscription plans, global geography, and viewing behavior.

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